Write legibly showing all the steps VERY clearly. (Total points = 100).

(1) (a) If a collection $\mathcal{G}$ is both a π-system and a λ-system, prove that it is a σ-algebra.
(b) Let $\Omega = \{a, b, c, d\}$ and $\mathcal{F} = \{\emptyset, \Omega, \{b\}, \{b\}^c, \{a, b\}, \{a, b\}^c\}$ be a collection. Is it a π-system? Is it a λ-system? Is it a σ-algebra? Justify your answer.
(c) Let $\Omega = I\!N$ and let $\mathcal{F}$ be the collection of subsets of Ω that are either finite or cofinite - t is shown in the text/notes that this is an algebra, but not a σ-algebra (do not need to prove this here). Prove that the counting measurwe is a valid measure on this *algebra.* (10+5×3+15 = 40 points)

(2) (a) Let $\mathcal{G}$ be a σ-algebra of sets of $\Omega \neq \emptyset$ and $\mathcal{G} \neq \mathcal{P}(\Omega)$. Take a $B \notin \mathcal{G}$ and define $\mathcal{G}_B = \{A \cap B : A \in \mathcal{G}\}$. Prove that $\mathcal{G}_B$ is a σ-algebra of subsets of B.
(b) Let $(\Omega, \mathcal{F})$ be a measurable space and for each $n \geq 1$, μ_n is a measure on this space. Argue that $\mu = \sum_{n=1}^{\infty} n^2 \mu_n$ is a measure on $(\Omega, \mathcal{F})$.
(c) Let $\mathcal{O}$ and $\mathcal{C}$ be the class of all open and closed sets of $\Omega = I\!R$ respectively and recall that $\mathcal{B}(I\!R) = \sigma\langle\mathcal{O}\rangle$ by definition. Argue that $\mathcal{B}(I\!R) = \sigma\langle\mathcal{C}\rangle$.
(3×10 = 30 points)

(3) Take $F(x) = (1 - e^{-x})\mathbf{I}_{(0,\infty)}(x), x \in I\!R$.
(a) Using the steps in Caratheodory Extension Theorem(s), show by starting from defining a measure μ_F on a suitable semi-algebra, how one can extend it to the Lebesgue-Stieltjes measure μ_F^* on $(I\!R, \mathcal{B}(I\!R))$.
[Write down the steps of the theorems carefully and state the results used in each step]
(b) Why is such a measure on $(I\!R, \mathcal{B}(I\!R))$ unique? What is the name of this measure?
(c) If $A = (0, 1)$, then compute $\mu_F^*(A)$.
(10×3=30 points)